

Course- B. Tech
Course Code- CSET343
Year- 4th Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Lab Assignment – 5

Objective: This assignment aims to provide hands-on experience with **multiclass** Classification, a classification problem with more than two target classes.

Note: You can work on either of the datasets given

Dataset: Dermatology

- Target Variable: Class (multiclass: 1-6, representing different skin diseases: 1=psoriasis, 2=seboreic dermatitis, 3=lichen planus, 4=pityriasis rosea, 5=chronic dermatitis, 6=pityriasis rubra pilaris).

Dataset: Vertebral

- Target Variable: Class (multiclass, representing 3 spinal conditions)

Task 1: Data Acquisition and Exploration

1. Download and load the dataset
2. Compute summary statistics (mean, median, standard deviation, etc.) for all features.
3. Check for missing values, duplicates, or outliers. Handle them
4. Visualize the data: Create histograms or box plots for key features, a correlation heatmap, and class distribution.
5. Discuss any imbalances in the class distribution and potential implications for model training.

Task 2: Data Preprocessing

- Encode the target variable.
- Split the data into training and testing sets (e.g., 80/20 split) using stratified sampling to maintain class balance.
- Standardize the features to ensure they are on the same scale, as gradient descent in some ML algos is sensitive to feature scales.

Task 3: Model Training and Basic Evaluation

- Implement a multiclass Machine Learning model (using 3-5 different ML algorithms such as KNN, RandomForest, LR, SVM etc.,)
- Evaluate and compare all the trained models on the test set using the following metrics:
 - Accuracy
 - Precision, Recall, and F1-score (report for both classes)
 - Confusion matrix (visualize it)
 - ROC curve and AUC score.
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- Interpret the results

